MILITARY SPECIFICATION

WATTMETERS, SWITCHBOARD

TYPE, 4-1/2-INCH

1. SCOPE

1.1 Scope.- This specification covers switchboard electrical indicating wattmeters including prefixes such as, kilo and mega.

1.2 Classification.-

1.2.1 Type designation.- The type designation of wattmeters shall be in the following form, and as specified (see 6.2).

MR	491	W	500	AC	WW	H
Component (1.2.1.1.)	Style (1.2.1.2)	Color scheme (1.2.1.3)	Full-scale value (1.2.1.4)	Kind of current (1.2.1.5)	Electrical units (1.2.1.6)	Special features (1.2.1.7)

1.2.1.1 Component.- Wattmeters are designated by the two-letter symbol "MR".

1.2.1.2 Style.- The style is designated by a three digit number which signifies the following characteristics:

1.2.1.2.1 First digit.- The first digit designates the nominal size and shape of flange on switchboard wattmeters as follows:

4 - 4-1/2-inch, rectangular flange.

1.2.1.2.2 Second digit.- The second digit designates the type of mounting and the nominal scale length in angular degrees, as follows:

9 - Flush mounting 260 degrees nominal scale.

1.2.1.2.3 Third digit.- The third digit designates either single phase or polyphase as follows:

1 - single phase
2 - polyphase (with 2 current coils and 2 voltage coils).

1 2.1.3 Color scheme.- The color scheme of the dial background, markings and pointer is designated by a single letter as follows:

B - Black dial background, white markings and pointer.
W - White dial background, black markings and pointer.

1.2.1.4 Full-scale value.-

1.2.1.4.1 Single-range wattmeters.- The full-scale value of single-range wattmeters is represented by three figures designating the maximum number of units indicated; when the full-scale value is less than three figures, zeros are inserted at the left to fill out three figures.

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Where the letter "R" is used in this group, between two figures, it represents a decimal point. Example: 1R5 represents 1.5.

1.2.1.4 2 Offset zero.- The full-scale deflection of zero-center or offset zero wattmeters is indicated by two numbers designating the numerical value of the end-scale points, with a letter between them designating the decimal value of the numbers, as follows:

D - tenths Example: 5D5 represents 0.5-0-0.5.
U - units Example: 1U9 represents 1-0-9.
T - tens Example: 2T2 represents 20-0-20.
H - hundreds Example: 5H5 represents 500-0-500.

1.2.1.5 Kind of current.- Two letters designate the kind of current for which the wattmeter is designated as follows:

AC - Alternating current (60 cycles per second, nominal frequency).
AD - Alternating current (60 cycles per second, nominal frequency), and direct current.
AE - Alternating current (800 cycles per second, nominal frequency).
AF - Alternating current (400 cycles per second, nominal frequency).

1.2.1.6 Electrical units.- Two letters designate the electrical units indicated by the wattmeters, as follows:

WW - Watts.
KW - Kilowatts.
GW - Megawatts.

1 2.1.7 Special features.- A single letter designates a special feature, as follows:

H - High-impact-shock resistant.

1.2.1.8 Special meters.- The four letters "SPEC" are used in place of the letters specified in 1.2.1.5 and 1.2.1.6 for wattmeters having special characteristics, such as adjusted resistance or special scale marks. These wattmeters are arbitrarily assigned serial numbers which are to be used in place of the three figures called for in 1.2.1.4.

1.2.1.9 Examples of type designations.-

1.2.1.9.1 Switchboard wattmeter.- The type designation MR491W300AC, KWH signifies:

(a) MR - Electrical-indicating wattmeter.
(b) 49 - 4-1/2-inch rectangular flange, switchboard wattmeter, flush-mounting, 250 degrees nominal scale.
(c) 1 - Single phase.
(d) W - White dial background with black markings and black pointer.
(e) 300 - Zero left, full-scale value of 300.
(f) AC - Designed to be energized by 60-cycle alternating current.
(g) KW - Electrical units in kilowatts.
(h) H - High-impact-shock resistant.

2. APPLICABLE DOCUMENTS

2.1 The follwing specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL
NN-B-591 - Boxes, Fiberboard, Wood-Cleated (for Domestic Shipment).
NN-B-621 - Boxes, Wood, Nailed and Lock-Corner.

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LLL-B-631 - Boxes, Fiber Corrugated (for Domestic Shipment).
LLL-B-636 - Boxes, Fiber, Solid (for Domestic Shipment).
PPP-B-601 - Boxes, Wood-Cleated-Plywood.

MILITARY

MIL-P-14 - Plastic-Materials, Molding, and Plastic-Parts, Molded; Thermosetting.
MIL-P-78 - Plastic-Material, Laminated, Thermosetting (for Designation Plates).
MIL-P-80 - Plastic Sheet, Acrylate Base, Clear, Transparent, Anti-electrostatic (for Indicator Dial Windows).
JAN-P-103 - Packaging and Packing for Overseas Shipment - Boxes, Wood-Cleated; Solid Fiberboard.
JAN-P-106 - Packaging and Packing for Overseas Shipment - Boxes, Wood, Nailed.
JAN-P-108 - Packaging and Packing for Overseas Shipment - Boxes, Fiberboard (V-Board and W-Board) Exterior and Interior.
MIL-P-116 - Preservation, Methods of.
MIL-B-857 - Bolts, Nuts, Studs, and Tap-Rivets (and Material for Same).
MIL-S-901 - Shockproof Equipment, Class HI (High-Impact), Shipboard Application, Tests for.
MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).
MIL-S-933 - Screws, Machine, Cap and Set; and Nuts.
MIL-I-1361 - Instruments, Electrical Measuring; Shunts, Resistors, and Transformers (Ship and Shore Use).
MIL-S-6872 - Soldering Process, General Specification for.
MIL-B-10377 - Boxes: Wood-Cleated, Veneer, Paper Overlaid.
MIL-L-10547 - Liners, Case, Waterproof.
MIL-I-15024 - Identification Plates, Information Plates, and Marking For Identification of Electrical, Electronic and Mechanical Equipment.

NAVY DEPARTMENT

General Specifications for Inspection of Material.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. - The following publications, of the issue in effect on date of invitation for bids, unless otherwise stated, form a part of this specification:

CONSOLIDATED CLASSIFICATION COMMITTEE

Consolidated Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

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3. REQUIREMENTS

3.1 Qualification.- Wattmeters furnished under this specification shall be a product which has been tested and has passed the qualification tests specified in 4.6 (see 6.3).

3.2 Definitions.-

3.2.1 Accuracy.- The accuracy is a number which defines the limit of error, expressed as a percentage of full-scale value unless otherwise specified herein.

3.2.1 Accuracy, rated.- The rated accuracy is a number assigned by the manufacturer to designate the accuracy classification of the wattmeter. This number, usually expressed as a percentage of full-scale value, is the limit which errors will not exceed when the wattmeter is used under reference conditions.

3.2.2 Current circuit.- The current circuit is that combination of conductors and windings of the wattmeter which carries the current of the circuit in which a given electrical quantity is to be measured, or a definite fraction of that current, or a current dependent upon it.

3.2.3 Damping.- Damping is the term applied to wattmeter performance to denote the manner in which the pointer settles to its steady indication after a change in the value of the measured quantity. The following general classes of damped motion are distinguished:

(a) Periodic, in which the pointer oscillates about the final position before coming to rest.
(b) Aperiodic, in which the pointer comes to rest without overshooting the rest position.

The point of change between periodic and aperiodic damping is called critical damping.

NOTE: A wattmeter is considered to be critically damped when overshoot is present but does not exceed an amount equal to one-half the initial-accuracy requirement of the wattmeter.

3.2.4 Damping factor.- The damping factor is the ratio of the steady deflection to the difference between maximum momentary deflection and steady deflection produced by a sudden application of constant electric power. All deflections are measured in angular degrees. In this specification, end-scale deflection is used as maximum momentary deflection.

NOTE: The damping factor of spring-controlled wattmeters with their normal accessories, if any, is measured as the ratio of the steady deflection in angular units to the differences between angular maximum momentary deflection and the steady deflection produced by a sudden application of sufficient constant electric power to an unenergized wattmeter to obtain end-scale deflection. This is the special case in which the deviation of the pointer from the position of equilibrium is the maximum value obtainable.

3.2.5 End-scale value.- The end-scale value of a wattmeter is the value of the actuating electrical quantity which corresponds to end-scale indication. When zero is not at one end or at the electrical center of the scale, the larger value is taken.

3.2.6 Error and correction.-

3.2.6.1 Error.- The error of indication is the difference between the indication and the true value of the quantity measured. It is the quantity which, when algebraically subtracted from the indication, gives the true value. A positive error denotes that the indication of the wattmeter is greater than the true value.

3.2.6.2 Correction.- The correction has the same numerical value as the error of indication, but the opposite sign. It is the quantity which, when algebraically added to the indication, gives the true value. If the symbols T, I, E, and C represent the true value, the indication, the error, and the correction, respectively, the following equations hold:

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EXAMPLE: A wattmeter indicates 112 watts when the wattage applied to its terminals is actually 110 watts.

THEN: Error = 112 - 110 = +2 watts.
Correction = 110 - 112 = -2 watts.

3.2.7 External-field influence.- The external-field influence is defined as the percentage change in the indication which is caused solely by an external field of an intensity of 5 oersteds, produced by a current of the same kind and frequency as that on which the wattmeter operates, with the most unfavorable phase and position of the external field.

3.2.8 Flush-mounting wattmeter.- In this specification, a flush-mounting wattmeter is a wattmeter designed so that, when mounted, the major portion of the wattmeter will extend behind the mounting panel.

3.2.9 Free indication.- Wattmeters shall be considered as indicating freely if the pointer of the wattmeter under test moves simultaneously with the pointer of the standard meter with which it is being compared, when the electrical energy supplied to both meters is gradually increased from zero to an amount sufficient to produce end-scale deflection of the wattmeter under test.

3.2.10 Frequency influence.- The frequency influence (in other than frequency meters) is defined as the percentage change in indication which is caused solely by a change of ± 10 percent from the rated frequency.

3.2.11 Full-scale value.- The full-scale value is equal to the largest value of the actuating electrical quantity which can be indicated on the scale; or, for wattmeters having the zero between the ends of the scale, the full-scale value is the arithmetic sum of the values of the actuating electrical quantity corresponding to the two ends of the scale.

3.2.12 Magnetic shielding.- Magnetic shielding is defined as limiting the effect of an external magnetic field of a designated strength to the value specified (see 3.27). Protection against magnetic fields may be inherent in the construction of the wattmeter, or it may be obtained by the use of a physical magnetic shield.

3.2.13 Mechanism.- The mechanism is the arrangement of parts for producing and controlling the motion of the indicating means. It includes all the essential parts necessary to produce the results, but does not include the case, cover, dial, or any parts, such as series resistors or shunts, whose function is to adapt the wattmeter to the quantity to be measured.

3.2.14 Wattmeter, electrical-indicating.- In this specification, an electrical-indicating wattmeter is defined as a measuring device which measures the value of the quantity under observation. The term "wattmeter" is used to include not only the wattmeter proper but, in addition, any necessary apparatus, such as shunts, shunt leads, resistors, reactors, condensers, or instrument transformers.

3.2.14.1 Wattmeter proper.- The wattmeter proper is the mechanism and the scale, built into the case, including all devices, such as resistors, shunts, and so forth, which are built into the case, or nonremovably attached to it.

3.2.15 Wattmeter-rating.- The wattmeter rating is a designation assigned by the manufacturer to indicate the operating limitations of the wattmeter. The end-scale marking of a wattmeter does not necessarily correspond to its rating.

3.2.16 Moving element.- The moving element of a wattmeter comprises those parts which move as a direct result of a variation in the electrical quantity which the wattmeter is measuring.

3.2.17 Normal operating position.- Vertical panel mounting is the normal operating position for wattmeters covered by this specification.

3.2.18 Overshoot.- Overshoot is the ratio of the overtravel of the pointer beyond its steady deflection to the change in steady deflection when the wattmeter is suddenly energized with a new

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constant value of the measured quantity. The overtravel and deflection are determined in angular measure and the overshoot is usually expressed as a percentage.

3.2.19 Position influence (balance).- The position influence is the maximum displacement of the pointer which is caused solely by a 60 degree inclination of the wattmeter from normal operating position in the direction which produces the most unfavorable conditions as to position.

3.2.20 Power consumption.- Power consumption is the electrical power consumed in the wattmeter to produce end-scale deflection of the wattmeter. Power consumption may be expressed in terms of units other than units of power.

3.2.21 Response time.- This term denotes the rapidity with which the pointer comes to rest after a change of the measured quantity. It may be measured as the time, in seconds, for the pointer to come to apparent rest after a change in the value of the measured quantity.

3.2.22 Scale, nominal.- The nominal scale for wattmeters shall be 250 degrees on the basis of angular scale length.

3.2.23 Scale division.- A scale division is the increment between the centers of two consecutive scale marks.

NOTE: The number ot scale marks is one more than the number of scale divisions. Example: 10 scale divisions require 11 scale marks.

3.2.23.1 Linear scales.- The total number of scale divisions is determined by dividing the total range by the smallest increment. Example: A 150 watt scale with the smallest increment of 5 watts is listed as having 30 scale divisions.

3.2.23.2 Nonlinear scales.- The total number of scale divisions is determined by dividing the total range by the smallest increment although, to avoid crowding scale marks, a portion of the scale may not be marked. Example: A 100 watt scale with the smallest increment of 2 watts, but with no marks between 0 and 10 watts, is listed as having 50 scale divisions.

3.2.24 Scale length.- The scale length is the length of the path described by the tip of the pointer in moving from one end of the scale to the other. In the case of knife-edge pointers and others extending beyond the scale division marks, the pointer shall be considered as ending at the outer end of the shortest scale-division marks. For multiple-scale wattmeters, the longest scale shall be used to determine the scale length.

3.2.25 Temperature influence.- Temperature influence is the percentage change in the indication caused solely by a difference in ambient temperature of plus or minus 10° C. from the reference temperature.

3.2.26 Voltage circuit.- The voltage circuit is that combination of conductors and windings of the wattmeter to which is applied the voltage of the circuit in which a given electrical quantity is to be measured, or a definite fraction of that voltage, or a voltage or current dependent upon it.

3.2.27 Voltage influence.- Voltage influence is the percentage change in the indication caused solely by a change of plus or minus 10 percent from the rated voltage.

3.2.28 Interelement Influence.- The interelement influence in polyphase wattmeters is the percentage change in the indication of the wattmeter which is caused solely by the action of the stray field of one element upon the other element.

3.3 Material.- The material for each part shall be as specified herein. When a definite material is not specified, a material shall be used which is adequate for the purpose intended and for meeting the requirements herein. Acceptance or approval of constituent material shall not be construed as a guaranty of the acceptance of the finished product.

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3.3 1 Plastic material.-

3.3.1.1 Molded thermosetting.- Molded thermosetting plastic material conforming to Specification MIL-P-14 shall be used, and shall be of the proper type for its intended use.

3.3.1.2 Laminated thermosetting.- Laminated thermosetting plastic materials used for identification plates shall conform to Specification MIL-P-78.

3.3.1.3 Plastic windows.- Plastic material used for wattmeter windows shall conform to Specification MIL-P-80.

3.3.1.4 Wattmeters shall conform to Specification MIL-E-917 with regard to the use of plastic material and other basic requirements of electric power equipment.

3.3.2 Fume-emitting material.- Material that emits deleterious or toxic fumes at 85° C. or lower shall not be used.

3.3.3 Coatings and finishes.- Coatings and finishes which will melt, crack, chip, or scale as a result of test specified in section 4 shall not be used.

3.4 Design and construction.-

3.4.1 Balancing.- The moving element shall be balanced by a means which will provide an easy method rebalancing, such as threaded lock- or split-nuts, or a wire helix moving along a cross arm and counterweight arm. The use of solder, or an adhesive such as shellac, to hold the balance weights or to function as a part of the balance weight, will not be acceptable.

3.4.2 Magnets.- If laminated or multiple-piece magnets are used, the laminations or components shall be securely fastened together so that they will not separate or loosen as a result of tests specified herein.

3.4.3 Threaded parts.- All threaded parts shall be in accordance with Handbook H28.

3.4.3.1 Bolts, nuts, and machine screws.- All bolts, nuts, and machine screws shall conform to Specification MIL-B-857 or MIL-S-933, as applicable.

3.4.3.2 Locking of screw-thread assemblies.- All screw-thread assemblies shall be made vibrationproof. When practicable, split-type lock washers or equivalent means shall be provided under all nuts.

3.5 Design and construction of component parts.-

3.5.1 Cases.- Wattmeter cases shall be made of molded thermosetting plastic material, metal, or other material as specified herein. The cases shall be designed and constructed with close-fitting joints to minimize the entrance of dust and moisture. The dimensions of switchboard meter cases shall be as specified (see 6.2).

3.5.1.1 Finish.- The portion of the case exposed to view from the front of the panel shall have a light gray finish having a Munsell rating of 2.5GY6.0/0.5. No nickel or other bright trimmings shall be used. Metal cases shall be rendered resistant to corrosion prior to the application of the final gray finish.

3.5.1.2 Window.- Wattmeters shall be provided with a window of glass, or other transparent material, free from discoloration, scratches, striae, electrostatic effect or other detrimental defects which may interfere in any way with normal reading of the wattmeter in daylight. Plastic windows, where used, shall be in accordance with Specification MIL-P-80. The window shall be secured by clamps, or by other suitable means. Cement alone is not considered sufficient for holding the window in place.

3.5.1.2.1 Scale visibility.- The scale and window shall be so designed that when the dial is vertical the entire scale of the meter can be easily read at any angle up to 45 degrees below the

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horizontal centerline on the vertical centerline, and 20 degrees to either side of the vertical centerline on the horizontal centerline, with general illumination coming from above and in front and at an angle 45 degrees from the horizontal centerline. Under these conditions, there shall be no objectionable shadow on the graduated scale.

3.5.2 Dial and pointer.

3.5.2.1 Dial.- The dial shall be made of stiff material firmly secured to its mounting and readily removable. The dial shall be of such material and so secured as to insure freedom from warping, fading, or other deterioration during the tests specified herein. Paper dials will be satisfactory only when adequately bonded to and backed by a metal support and when they meet all tests specified herein without fading, discoloring, warping, or other deterioration.

3.5.2.1.1 Scale.- Full scale ranges shall be as follows:

Watts [1]	Kilowatts	Megawatts
500 1000	As required	As required

[1] For zero center wattmeters the full scale range shall be as required.

3.5.2.1.1.1 Length.- The scale length shall be as specified (see 3.27).

3.5.2.1.1.2 Divisions.- The value of each scale division shall, whenever practicable, be either one, two, or five of the units measured or any decimal multiple or submultiple of these numbers. In the case of multiple-range wattmeters or of wattmeters for other specialized uses, exceptions to this requirement may be necessary, but shall be avoided when reasonably possible. The angle subtended by a scale division shall be not less than 1 degree on switchboard wattmeters.

3.5.2.2 Pointer.- The wattmeter shall have a pointer so formed as to permit accurate readings at the usual distance from which the wattmeter is viewed, and approximate readings up to the distance of legibility of the scale markings. The pointer shall be of rigid construction and suitable for the intended application, and shall not become damaged or distorted when subjected to any of the tests specified herein.

3.5.2.2.1 Pointer clearance (parallax).- Where the wattmeter is not provided with means for avoiding errors in reading due to parallax, the clearance between the pointer index and the graduated scale shall not exceed 0.10 inch, nor be less than 0.03 inch.

3.5.3 Means of adjustment.-

3.5.3.1 Accuracy adjustment.- Wattmeters shall be provided with practical shop means of adjustment to accuracy. Charging of the magnet and magnetically treating it to obtain accuracy, or adjusting the internal resistor, shall be considered as meeting this requirement.

3.5.3.2 Zero adjustment.- It is preferred that the zero adjuster be capable of rotating through 360 degrees without damage to any part of the wattmeter and without becoming inoperative. If the zero adjuster is not designed for 360 degrees rotation, it shall be capable of withstanding a torque of 10 inch-ounces without damage to the zero-adjuster or to any part of the meter.

3.5.4 Terminals and connections.-

3.5.4.1 Terminal studs.- Terminal studs shall be firmly anchored to the base of the wattmeter in such a manner as to prevent turning or movement sufficient to cause damage to internal construction. Each terminal stud shall be furnished with the necessary nuts or screws, and washers.

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3.5.4.2 Connection diagrams.- Wattmeters shall have a print showing the wiring diagram of connections with or without transformers, resistances, and reactances for three-phase three-wire systems and two-phase four-wire systems, together with instructions attached permanently or sufficiently secured to prevent loss during shipment.

3.5.5 Self-contained limitations.- Unless otherwise specified in the contract or order, wattmeters shall be self-contained with 20 ampere current circuit and external accessories may be used for all voltage ranges.

3.5.6 Mounting.- Switchboard wattmeters shall be mounted by means of mounting bolts or studs. The necessary mounting bolts (screws), nuts, and lock washers shall be supplied with wattmeter. The wattmeters shall mount on a panel drilled in accordance with the specified panel-drilling plan shown on figure 1.

3.6 External accessories.- External shunts, shunt leads, resistors, and transformers shall be furnished when specified (see 6.2). External accessories shall conform to Specification MIL-I-1361.

3.7 Calibration.-

3.7.1 Position.- Wattmeters shall be calibrated in the position in which they are normally used, and shall meet the specified accuracy requirements in such position at the reference temperature (25° C.).

3.7.2 Mounting panel.- Wattmeters shall meet the specified accuracy requirements when mounted on a nonmagnetic panel and also on a magnetic-steel panel 1/8 inch thick.

3.7.3 Frequency.- Wattmeters shall be calibrated at the following frequencies:

Kind of current	Frequency (c.p.s)
AC	60
AD	60
AF	400
AE	800

3.8 Position influence (balance).- When spring-controlled switchboard wattmeters are tilted 60 degrees from their normal mounting position, in any direction, the pointer tip shall not deviate from the zero mark more than the amount specified (see 3.27 and 4.6.3).

3.9 Initial accuracy.- At the reference temperature, the initial accuracy, expressed as a percentage of the full-scale value, shall be within the limits specified (see 3.27 and 4.6.4).

3.10 Damping factor.- When tested in accordance with 4.6.5, the damping factor shall be not less than the value specified (see 3.27).

3.11 Response time.- When tested in accordance with 4.6.6, the response time shall not exceed the value specified (see 3.27).

3.12 Power consumption.- The power consumption shall not exceed the value specified (see 3.27 and 4.6.7).

3.13 Frequency range.- Alternating-current wattmeters having a nominal frequency of 60 cycles per second shall operate over the range specified, and the change in indication from the reading at the nominal frequency shall not exceed the value specified (see 3.27 and 4.6.8).

3.14 Frequency influence.- The frequency influence shall be not greater than the value specified (see 3.27 and 4.6.9).

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3.15 External-field influence.- Wattmeters shall have sufficient magnetic shielding so that the effect of an external magnetic field of 5 oersteds will not cause a change in indication greater than the value specified (see 3.27 and 4.6.10).

3.16 Heat effect at 65° C.- Wattmeters shall indicate freely (see 3.2.9) during and after the test specified in 4.6.11. The heat effect at 65° C. (the difference between corresponding readings of steps 9 and 11 of table VII), and the permanent change in indication (the difference between corresponding readings of steps 1 and 13 of table VII), shall not exceed the values specified (see 3.27).

3.17 Temperature influence.- The temperature influence shall not exceed the value specified (see 3.27 and 4.6.12).

3.18 Low-temperature exposure.- During exposure to a temperature of 0° C., wattmeters shall indicate freely (see 3.29). The change in indication at the low temperature, and the permanent change in indication, shall not exceed the values specified (see 3.27 and 4.6.13).

3.19 Overload.-

3.19.1 Momentary overload.- The mechanism of a.c. wattmeters shall be sufficiently rugged to withstand, without mechanical or thermal injury, a series of momentary overloads as specified in 4.6.14.1. As a result of this test, it shall not be necessary to make any repairs or adjustments other than resetting the pointer to zero, through the external zero-adjuster, and the permanent change in indication shall not exceed the value specified (see 3.27).

3.19.2 Sustained overload.- When tested as specified in 4.6.14.2, the permanent change in indication, and the temporary and permanent zero shifts shall not exceed the value specified (see 3.27).

3.20 Dielectric stress.- When tested in accordance with 4.6.15, wattmeters shall withstand a voltage of 2,600 volts alternating current without damage or flashover.

3.21 Insulation resistance.- The insulation resistance shall be not less than 20 megohms (see 4.6.16).

3.22 Vibration.- When wattmeters are tested in accordance with 4.6.17, no screws, bearings, pivots, or other parts shall be so loosened or damaged as to cause a change in indication greater than the value specified (see 3.27).

3.22.1 Friction effect.- After completion of the vibration test specified in 4.6.17, the friction effect shall not exceed the value specified (see 3.27 and 4.6.17.1).

3.23 Humidity resistance.- After the test specified in 4.6.18, wattmeters shall operate freely, and the permanent change in indication shall not exceed the value specified (see 3.27).

3.24 Shock (high-impact).- When high-impact-shock resistant wattmeters are tested in accordance with 4.6.19, no screws, bearings, pivots, windows, or other parts shall become loosened, cracked, or unduly damaged. The permanent change in indication based on calibration made after resetting the pointer to zero shall not exceed the value specified (see 3.27).

NOTE: The design of high-impact-shock-resistant wattmeters shall be such that disassembly and proper reassembly prior to the shock test will not impair the shock resistance of the wattmeter.

3.25 Marking.- Wattmeters shall be permanently and legibly marked with the applicable information specified in 3.25.1.

3.25.1 Dials.- Dial markings shall be sharply defined and visible from the front of the case. These markings shall not, however, be so conspicuous as to distract attention from the scale markings. Unless otherwise specified in the contract or order, the scale markings and numerals,

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and items (b) and (c), below, when applicable, shall be marked on the dial; items (a), and (d) through (j) shall be marked on the dial, on the front of the wattmeter, or on an attached identification plate conforming to Specification MIL-I-15024.

(a) Type designation (see 1.2.1).
(b) Designation of the quantity measured. (Watts, kilowatts or megawatts.)
(c) The words "alternating current", or their abbreviations.
(d) Manufacturer's name or trade-mark (or symbol), or both.
(e) Manufacturer's type or model number, or both.
(f) The electrical quantity producing full-scale deflection when this differs from the full-scale value. Example: FS = 500 watts.
(g) External accessories as required. Example: Use with 50 mv shunt.
(h) Ratio of the appropriate current transformer, expressed as: 1000: 5 or 1000/5 or 200: 1 or 200/1.
(i) Ratio of the appropriate potential (voltage) transformer, expressed as: 3300: 110 or 3300/110 or 30: 1 or 30/1.
(j) Loop resistance of leads for external shunt of wattmeter, when the lead resistance affects accuracy (see 6.2).

3.26 Interelement influence.- The interelement influence shall not exceed the value specified (see 3.27 and 4.6.20).

3.27 Design limits.- The limits for accuracy shall be as follows:

Item	Limits
(a) Scale length	6.5 inches minimum
(b) Position influence (balance)	1 percent maximum change
(c) Initial accuracy	1 percent maximum error, (1-1/2 percent maximum error for HI shock meters)
(d) Damping factor	3 minutes (overshoot, 20 percent maximum)
(e) Response time	2.5 seconds maximum
(f) Power consumption	Potential system 7.5 volts ampere maximum per element at 120 volts for nominal 120 volt rating. For higher nominal voltage ratings the maximum volt ampere shall be increased proportionately. Current system 2.5 volt ampere maximum per element with current applied in phase with the potential to produce end scale deflection while the potential system is energized at rated voltage, 60 cycles. In a 2 element system this is determined with all current coils similarly energized.
(g) Frequency influence	0.5 percent maximum change for 60 cycles; 2 percent maximum change for 400 and 800 cycles.
(h) Voltage influence	0.5 percent maximum with plus or minus 10 percent voltage variation from nominal voltage rating.
(i) Power factor influence	1 percent maximum from 1.0 to 0.5 leading or lagging power factor.

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Item	Limits
(j) External-field influence	3 percent maximum change in 5 oersted field.
(k) Heat effect at 65° C.	1 percent maximum change at 65° C; 0.5 percent maximum permanent change.
(l) Temperature influence	.75 percent maximum change.
(m) Low temperature exposure	1.88 percent maximum change at 0° C; 0.5 percent permanent change.
(n) Sustained overload	1 percent maximum temporary zero shift. 0.5 percent maximum permanent zero shift. 0.33 percent maximum change in indication.
(o) Vibration	1 percent maximum change.
(p) Friction effect	2 percent maximum change.
(q) Humidity resistance	2 percent maximum change.
(r) Shock (HI impact)	5 percent maximum change.
(s) Interelement influence	0.5 percent maximum change.

NOTE:
1. Percentages are based on full scale length unless otherwise indicated.

3.28 Soldering. - Soldering shall be in accordance with Specification MIL-S-6872.

3.29 Workmanship. - Wattmeters shall be manufactured and processed in a careful and workmanlike manner in accordance with good design and sound practice. The interiors of wattmeters shall be free from metal filings, grease or oil, foreign material, dust, or other loose particles which might interfere with the normal operation of the wattmeters.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of tests. - The methods of sampling, inspection, and tests conducted on wattmeters shall be classified, as follows:

(a) Qualification tests (see 4.3).
(b) Inspection tests (see 4.5).

4.2 Test conditions and general instructions. -

4.2.1 Atmospheric conditions. -

4.2.1.1 Qualification tests. - Unless otherwise specified herein, all qualification tests shall be made at ambient temperature of 23 to 27° C., at existing barometric pressure, and with relative humidity of 40 to 60 percent.

4.2.1.2 Inspection tests. - Unless otherwise specified herein, all inspection tests shall be made at ambient temperature of 20 to 30° C., at existing barometric pressure, and with relative humidity less than 80 percent.

4.2.2 Reference temperature. - The reference temperature shall be 25° C.

4.2.3 Position. - Unless otherwise specified herein, tests performed on wattmeters, including all tests requiring scale readings, shall be performed with the wattmeter in the position in which it is normally used.

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4.2.4 Examination of pivots and bearings.- Examination of pivots and bearings for evidence of corrosion, damage from vibration, shock, and other tests, shall be made through a microscope having a magnification of 40 to 100 diameters.

4.2.5 Tapping.- When taking wattmeter readings, either the wattmeter or its support shall be tapped lightly, except when ascertaining the effect of friction after the vibration test.

4.2.6 Parallax.- Care should be exercised to avoid the effect of parallax when taking wattmeter readings.

4.2.7 Resetting to zero.- Unless otherwise specified herein, wattmeters provided with an external zero-adjuster may be set to zero at the completion of each test.

4.2.8 Thermal stability.- For all tests, care shall be taken to obtain thermal stability at the specified temperature prior to reading the wattmeter.

4.2.9 Number of scale readings.- Unless otherwise specified herein, readings on wattmeters shall be taken at six approximately equidistant points on the scale including the zero and end-scale readings.

4.2.10 Standard meters.- Standard meters shall be used to check the indications of wattmeters submitted for test. Accuracies and tolerances specified are absolute, and readings of standard meters should be corrected before determining compliance.

4.3 Qualification tests at a Government laboratory.- Qualification tests shall be conducted at a Government laboratory designated by the Bureau of Ships. These tests shall consist of the tests specified in 4.6.

4.3.1 Samples.- Sample wattmeters submitted for qualification approval shall be representative of the manufacturer's normal production and shall be taken, preferably, from a current production lot. The type designation and number of specimens required to be submitted for each kind of wattmeter shall be as specified in table I. Sample wattmeters which are used with external shunts shall be supplied with shunt leads of 0.065 ± 0.010-ohm resistance. Where 400- and 800-cycle wattmeters are similar in basic construction to the submitted 60-cycle wattmeter, only one additional specimen of each type is required. All qualification tests shall be performed on this specimen except the tests for external-field influence, vibration, humidity resistance, and shock (high-impact). If the basic design is different, a full number of specimens shall be submitted, and all qualification tests shall be performed.

Table I - Samples for qualification tests of wattmeters.

Kind of Wattmeter		Specimens		Ranges covered (end scale)
		Type designation	Number required	
4-1/2 inch, 250 degrees nominal scale, high-impact-shock resistant	A.C. single phase	MR491W500ACWWA	3	All ranges
4-1/2 inch, 250 degrees nominal scale, high-impact-shock resistant	A.C. polyphase	MR492W1000ACWWA	3	All ranges
4-1/2 inch, 250 degrees nominal	A.C. single phase	MR491W500ACWW	2	All ranges 500 W and above

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Table I - Samples for qualification tests of wattmeters. (cont'd)

Kind of Wattmeter		Specimens		Ranges covered (end scale)
		Type designation	Number required	
4-1/2 inch, 250 degrees nominal scale	A.C. polyphase	MR492W500ACWW	2	All ranges 500 W and above

NOTE: When approval is desired for 400- or 800-cycle meters, the range of the specimens submitted shall be the same as that specified for 60-cycle meters. The number of specimens required shall be in accordance with 4.3.1.

4.3.2 Test routine.- Sample wattmeters shall be subjected to the applicable qualification tests shown in table II, in the order listed.

Table II - Qualification tests.

Test	Requirement paragraph	Test paragraph
Visual and mechanical inspection (without opening case)	(see table VI)	4.6.1
Zero adjustment	3.5.3.2	4.6.2
Position influence (balance)	3.8	4.6.3
Initial accuracy	3.9	4.6.4.1
Damping factor	3.10	4.6.5
Response time	3.11	4.6.6
Power consumption	3.12	4.6.7
Frequency range	3.13	4.6.8
Frequency influence	3.14	4.6.9
External-field influence	3.15	4.6.10
Heat effect at 65° C.	3.16	4.6.11
Temperature influence	3.17	4.6.12
Low-temperature exposure	3.18	4.6.13
Overload	3.19	4.6.14
Dielectric stress	3.20	4.6.15
Insulation resistance	3.21	4.6.16
Vibration	3.22	4.6.17
Humidity resistance (1 specimen)	3.23	4.6.18
Shock (high-impact) (2 specimens)	3.24	4.6.19
Interelement influence	3.26	4.6.20
Visual and mechanical inspection (external and internal)	(see table VI)	4.6.1

4.3.3 Failures.- One failure will be permitted on a sample of wattmeters under the following conditions:

(a) The failure is limited to one of the following tests:
1. Position influence (balance).
2. Initial accuracy.

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3. Damping factor.
4. Response time.

(b) The magnitude of the failure is limited to 1-1/2 times the specified tolerance for the initial-accuracy test, and 2 times the specified tolerance for the other three tests.

4.3.3.1 Where material specifications are specified, the manufacturer shall certify that the materials used are of the proper grade and are in accordance with the designated specification.

4.4 Sampling.-

4.4.1 Lot.- All wattmeters of the same design offered for delivery at one time shall be considered a lot.

4.4.2 Sampling for group A tests.- Statistical sampling for group A tests shall be in accordance with Standard MIL-STD-105. Major and minor defects shall be as defined in Standard MIL-STD-105 and the acceptable quality level (AQL) shall be as specified in 4.5.1.

4.4.3 Sampling for group B tests.- Group B tests shall be performed on sample units that have been subjected to and have passed the group A tests, unless the Government considers it more practical to select a separate sample from the lot for group B tests, in which case the sampling procedure shall be in accordance with inspection level L7 of Appendix to Standard MIL-STD-105. The acceptable quality level (AQL) shall be 6.5 percent. All sample units shall be selected without regard to their quality.

4.4.4 Sampling for group C tests.- Sampling for group C tests shall be performed on two sample wattmeters from each lot regardless of their quantity.

4.5 Lot acceptance inspection tests.-

4.5.1 Group A tests.- Each of the sample wattmeters selected in accordance with 4.4.2 shall be subjected to each of the tests specified in table III, in the order shown. Tests shall be performed at the place of manufacture by or under the supervision of the Government inspector.

Table III - Group A tests.

Test	Requirement paragraph	Test paragraph	AQL (percent defective)	
			Major	Minor
Visual and mechanical inspection (without opening case)	(See table VI)	4.6.1	1.0	4.0
Zero adjustment	3.5.3.2	4.6.2	1.0	---
Position influence (balance)	3.8	4.6.3	1.0	---
Initial accuracy (2 points)	3.9	4.6.4.2	1.0	---

4.5.2 Group B tests.- Samples selected in accordance with 4.4.3 shall be subjected to each of the tests specified in table IV in the order shown. Tests shall be performed at the place of manufacture by or under the supervision of the Government inspector. Failure to conform to this specification for any group B test shall be counted as a defect. If the number of defective units exceeds the applicable acceptance number, the lot shall be rejected. If a lot is rejected due to failure to pass group B tests, the manufacturer may rework the lot or screen out defectives and resubmit it for Government inspection. Resubmitted lots shall be kept separate from new lots. The resubmitted lot shall be inspected, using inspection level L7 of appendix to Standard MIL-STD-105. Sample units which have been subjected to and have passed the group B tests shall be subjected again to the initial-accuracy test.

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Table IV - Group B tests.

Test	Requirement paragraph	Test paragraph
Initial accuracy (7 points)	3.9	4.6.4
Frequency range	3.13	4.6.8
Frequency influence	3.14	4.6.9
External field influence	3.15	4.6.10
Inter-element influence	3.26	4.6.20
Dielectric stress	3.20	4.6.15
Insulation resistance	3.21	4.6.16
Damping factor	3.10	4.6.5
Response time	3.11	4.6.6
Power consumption (loss)	3.12	4.6.7

4.5.3 Group C tests.- Samples selected in accordance with 4.4.4 shall be subjected to the group C tests specified in table V, in the order shown. Tests shall be performed at the place of manufacture by or under the supervision of the Government inspector. Any defective units shall be cause for rejection. If a sample fails group C tests, the manufacturer shall take corrective action on the processes and all units of product which were manufactured with the same conditions, materials, processes, and so forth, and are considered subject to the same test failure. Government inspection may be discontinued, at the option of the Government inspector, until a sample produced by the corrected process has been subjected to and has passed the group C tests failed by the preceding sample. Units of product subsequently submitted for Government inspection shall incorporate the corrective action, unless this requirement is waived temporarily by the bureau or agency concerned.

Table V - Group C tests.

Test	Requirement paragraph	Test paragraph
Heat effect at 65° C.	3.16	4.6.11
Temperature influence	3.17	4.6.12
Low-temperature exposure	3.18	4.6.13
Overload:		
Momentary	3.19.1	4.6.14.1
Sustained	3.19.2	4.6.14.2

4.6 Test procedures.-

4.6.1 Visual and mechanical inspection.- Wattmeters shall be inspected to determine compliance with table VI.

Table VI - Visual and mechanical inspection

Requirement	Paragraph	Defect classification
Material	3.3 to 3.3.3, incl.	Major
Design and construction	3.4	Minor
Cases	3.5.1 to 3.5.1.2.1, incl.	Major
Dial and pointer	3.5.2	Minor

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Table VI - Visual and mechanical inspection. (Contd)

Requirement	Paragraph	Defect classification
Means of adjustment	3.5.3	Minor
Terminals and connections	3.5.4	Minor
Mounting	3.5.6	Major
Marking	3.25	Minor
Workmanship	3.28 and 3.29	Major

4.6.2 Zero adjustment. - The range of the zero adjustment above and below the center of the zero mark shall be determined. Record shall be made as to whether or not the zero-adjuster may be rotated through 360 degrees without damage to the wattmeter or if the design precludes a 360 degree rotation whether or not the zero adjuster is capable of withstanding a torque of 10 inch-ounces without damage to the zero adjuster or to any part of the meter (see 3.5.3.2).

4.6.3 Position influence (balance). - The pointer shall be set to the center of the zero scale division with the wattmeter in the normal operating position. The wattmeter shall then be tilted 60 degrees from this position in any direction, and the deviation of the pointer noted (see 3.8).

4.6.4 Initial accuracy. - Wattmeters shall be checked for accuracy as specified in 4.6.4.1 or 4.6.4.2, as applicable.

4.6.4.1 Qualification test. - After preliminary conditioning in the test circuit for 30 minutes at two-thirds scale deflection, readings shall be taken and compared with corresponding readings of the standard meter. The test shall be made with the wattmeter at a temperature of 25° C. (see 3.9 and 4.2.9).

4.6.4.2 Inspection test. - The inspection test of initial accuracy shall be performed under the test conditions specified in 4.2 without preliminary conditioning (see 3.9).

4.6.5 Damping factor. - Electric power sufficient to produce momentary end-scale deflection shall be applied to the wattmeter. The maximum momentary deflection shall be observed and recorded. After the pointer has come to rest, the steady deflection shall be observed and recorded. When making this test, the circuit resistance shall be not less than 100,000 ohms and such that no additional damping is added to the system. The damping factor shall be taken as the ratio of the steady deflection (in angular units) to the difference between the maximum momentary deflection and the steady deflection in angular units. Example: If the maximum momentary deflection is 90 angular degrees and the steady deflection is 75 angular degrees, the difference bewteen the two is 15 degrees, and the damping factor (the ratio of 75 degrees to 15 degrees) is 5 (see 3.10).

4.6.6 Response time. - Steady electric power sufficient to produce momentary end-scale deflection shall be applied to the wattmeter. The length of time, in seconds, required for the pointer to come to apparent rest shall be recorded. The pointer shall be considered as having come to apparent rest when it has obtained the actual point of rest within one-half the initial-accuracy requirement. The test shall be repeated five times, and the average length of time recorded for all measurements shall be considered as the response time of the wattmeter (see 3.11).

4.6.7 Power consumption. - The power consumption shall be measured at end-scale deflection and recorded in terms of current, voltage, voltamperes, or ohms, as applicable (see 3.12).

4.6.8 Frequency range. - Unless otherwise specified in the contract or order, wattmeters having a nominal frequency of 60 cycles per second shall be tested for satisfactory operation at the extreme operating frequencies specified (see 3.13).

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4.6.9 Frequency influence.- Wattmeters shall be calibrated at nominal frequency, then at frequencies 10 percent above and 10 percent below the nominal frequency, and the change in indication determined (see 3.14).

4.6.10 External-field influence.- The wattmeter shall be placed at the center of a four-turn coil, 50 centimeters in diameter, carrying a current of 50 amperes of the same kind and frequency as that which actuates the wattmeter mechanism, and which produces a field of approximately 5 oersteds at the center of the coil. Readings shall be taken with the wattmeter at various positions in the center of the field (see 3.15).

NOTE: It is the intent to specify a test coil which will produce a uniform field of approximately 5 oersteds through the case of the wattmeter. Therefore, coils of other sizes are equally suitable if an equivalent electrical field is produced.

4.6.11 Heat effect at 65° C.- Wattmeters shall be maintained at reference temperature for a sufficient time (not less than 2 hours) to attain thermal stability, and shall then be subjected to the three successive cycle of temperature change listed in table VII. During this test, wattmeters shall indicate freely (see 3.2.9) at all parts of the scale (see 3.16).

Table VII - Procedure for determining heat effect at 65°C.

Cycle number	Period (hours)	Step number	Procedure
		1	Take initial reading at reference temperature.
	16	2	Place meter in chamber at 65° C. and maintain that temperature for the remainder of that period.
		3	Read meter while at 65° C. at the end of step 2.
	8	4	Allow chamber and meter to cool down to the reference temperature during this period.
1		5	Read meter while at reference temperature at the end of step 4.
	16	6	Raise temperature of chamber and meter as rapidly as possible to 65° C. and maintain that temperature for the remainder of the period.
2		7	Read meter while at 65° C. at the end of step 6.
	8	8	Allow chamber and meter to cool down to the reference temperature during this period.
		9	Read meter while at reference temperature at the end of step 8.
	16	10	Raise temperature of chamber and meter as rapidly as possible to 65° C. and maintain that temperature for the remainder of the period.
		11	Read meter while at 65° C. at the end of step 10.
3	8	12	Allow chamber and meter to cool down to the reference temperature during this period.
		13	Read meter while at reference temperature at the end of step 12.

4.6.12 Temperature influence.- Wattmeters shall be tested at temperature 20° C. above and below the reference temperature. Readings shall be taken at not less than four points on the scale, one of which is at or near the maximum scale value, and the temperature influence computed. The temperature influence shall be considered one-half the percentage change in indication caused by the 20° C. departures from the reference temperature. If the influences above and below the reference temperature are not equal, the greater value shall be considered the temperature influence (see 3.17).

4.6.13 Low-temperature exposure.- Wattmeters shall be exposed to a temperature of 0° C. for 16 hours, and at the end of the period wattmeter readings shall be taken. It shall be observed whether or not the wattmeter indicated freely at all parts of the scale while at 0° C. Wattmeters

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shall then be maintained at room temperature for sufficient time to attain thermal stability, and meter readings repeated. The permanent change in indication resulting from the low-temperature exposure shall be noted (see 3.18).

4.6.14 Overload.-

4.6.14.1 Momentary overload.- The current circuits of wattmeters shall be subjected to 10 applications of current equal to 10 times end-scale value. The current shall be applied nine times for a period of one-half second each time, with 1-minute intervals between successive applications, then followed by one application for a 5-second period (making a total of 10 applications during an elapsed period of 9 minutes, 9-1/2 seconds), during which time current is applied for a total of 9-1/2 seconds. (If the wattmeter is provided with a voltage circuit, normal voltage shall be applied to the wattmeter during the whole period of the test.) After 1 hour at reference temperature following the above test, the pointer shall be reset to zero and the change in indication determined. This test should not be applied to external shunts (see 3.19.1).

4.6.14.2 Sustained overload.- Wattmeters shall be subjected for 8 hours to an application of energy 20 percent greater than end-scale value. Immediately after the load is withdrawn at the end of the 8-hour period the temporary zero shift shall be determined. Sixteen hours after completion of the 8-hour overload period, the permanent zero shift shall be noted. The pointer shall then be set to zero and the permanent change in indication determined (see 3.19.2).

4.6.15 Dielectric stress.- When performing this test, wattmeters with nonmetallic cases shall be mounted on a metallic panel. All meters shall be subjected to a voltage of 2,600 volts r.m.s. applied between all external live parts of the wattmeter and:

(a) The metallic platform or panel on which the wattmeter is placed or mounted, or the wattmeter case, if the case is metallic.
(b) The zero-adjuster; and range selector switch, when applicable.
(c) All exposed metal parts not a live part of the wattmeter.

The voltage shall be nominal 60 cycles per second in frequency and shall approximate, as closely as possible, a true sine wave in form. The voltage shall be applied continuously for a period of 60 seconds. The test voltage shall be measured by the voltmeter method whereby the measuring meter derives its voltage from the high-potential circuit, either directly or by means of a voltmeter coil placed in the testing transformer, or by means of an auxiliary ratio transformer. To avoid voltage surges, the applied voltage shall be raised to its full value gradually, and upon completion of the 60-second test period shall be gradually reduced. The test shall be made using high-potential test prods having not less than 1/8-inch radius tips. Any damage or flash-over shall be noted (see 3.20).

4.6.16 Insulation resistance.- Wattmeters shall be placed on a metallic platform or mounted on a metallic panel as specified in 4.5.15. Using a potential of 500 volts direct current, the insulation resistance shall be determined between all external live parts and the metallic platform or panel on which the wattmeter is placed or mounted, or the wattmeter case, if the case is metallic (see 3.21).

NOTE: For inspection tests of meters furnished with external accessories as an integral part, the complete wattmeter (meter with accessories) shall be tested.

4.6.17 Vibration.- Wattmeters shall be tested in accordance with type I tests of Standard MIL-STD-167. (See 3.22.)

4.6.17.1 Friction effect.- Upon completion of the vibration test, the friction effect shall be determined, as follows: Untapped ascending and descending readings shall be taken at five points distributed over the scale, using the same scale points for ascending and descending readings. The difference between corresponding ascending and descending readings as observed on the standard meter shall be determined (see 3.22.1).

4.6.18 Humidity resistance.- Wattmeters readings shall be taken at reference temperature. The wattmeter shall then be placed in a humidity chamber and kept for 6 hours at a temperature

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of 65° C. with relative humidity of 90 percent. Sufficient electrical energy shall be supplied to the wattmeter to maintain two-thirds end-scale deflection. At the end of the 6-hour period, the heat shall be shut off and the wattmeter de-energized. The wattmeter shall then be permitted to cool for 18 hours in this atmosphere, the relative humidity of which rises as the temperature is reduced. (Precaution shall be taken to prevent dripping on the wattmeter.) A temperature of 35° C. or lower, but not less than 25° C., shall be attained during the cooling period. The complete cycle shall then be repeated. Upon completion of the second cycle, the wattmeter shall be removed from the humidity chamber and permitted to dry for a period of 24 hours at room temperature, with relative humidity less than 60 percent. Wattmeter readings shall then be taken at reference temperature. The wattmeter window shall then be removed and the wattmeter, with window removed, subject to two additional cycles. The wattmeter shall then be removed from the humidity chamber and permitted to dry for a period of 24 hours at room temperature, with relative humidity less than 60 percent. Wattmeter readings shall then be taken at reference temperature. After the drying period, a maximum of 10 momentary end-scale deflections will be permitted prior to the final reading. Only one wattmeter of the sample shall be subject to this test (see 3.23).

4.6.19 Shock (high-impact).- High-impact-shock-resistant wattmeters shall be tested in accordance with type C of Specification MIL-S-901 for lightweight equipment, when mounted on a plate as shown on figure 6D of that specification. Two wattmeters of the sample, which were not subjected to the humidity-resistance test, shall be used. During the test, one of the wattmeters shall be energized to approximately one-half scale deflection. A total of nine blows shall be applied. Three blows shall be applied parallel to each of the three principal axes of the wattmeters, the three blows for each direction to be with heights of hammer drop 1 foot, 3 feet, and 5 feet. The pointer shall be reset to zero after each blow. At the completion of the shock test, the wattmeters shall be checked for calibration errors, then disassembled and carefully examined for damage or misalinement of parts (see 3.24).

4.6.20 Interelement influence.- Polyphase wattmeters shall be energized with rated voltage and current in both elements such as to give end-scale deflection. Both current and voltage in one element shall then be reversed and one-half the difference in the readings in percent shall be recorded as the interelement influence (see 3.26).

4.7 Inspection procedures.- For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

5. PREPARATION FOR DELIVERY

5.1 Packaging.-

5.1.1 For domestic shipment - immediate use.- Packaging shall be in accordance with the manufacturer's commercial practice.

5.1.2 For domestic shipment and storage or overseas shipment.- Each wattmeter shall be individually cushioned and unit packaged in accordance with method IIb of Specification MIL-P-116.

5.2 Packing.-

5.2.1 For domestic shipment - immediate use.- Wattmeters packaged as specified in 5.1.1 shall be packed in a manner to insure acceptance and safe delivery at destination. Containers shall comply with the Consolidated Freight Classification Rules or other carrier regulations applicable to the mode of transportation.

5.2.2 For domestic shipment and storage.- Wattmeters packaged as specified in 5.1.2 shall be packed in wood-cleated fiberboard, wood-cleated plywood, wood-kraft veneer board, nailed wood, or fiberboard boxes conforming to Specification NN-B-591, PPP-B-601 (domestic type), MIL-B-10377 (domestic type), NN-B-621, or the special requirements of Specification LLL-B-631 or LLL-B-636. Boxes shall be closed in accordance with the applicable box specification. The gross weight of wood or wood-cleated containers shall not exceed 200 pounds, of fiberboard containers 70 pounds.

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5.2.3 For overseas shipment.- Wattmeters packaged as specified in 5.1.2 shall be packed in wood-cleated fiberboard, wood-cleated plywood, wood cleated kraft-veneer board, nailed wood or fiberboard boxes conforming to Specification JAN-P-103, PPP-B-601 (overseas type), MIL-B-10377 (overseas type), JAN-P-106, or JAN-P-108 (exterior grade). Boxes shall be sealed, closed and strapped in accordance with the appendix of the applicable box specification. Wood of wood-cleated containers shall be lined with a sealed water-proof case liner conforming to Specification MIL-L-10547. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds, of fiberboard boxes 70 pounds.

5.3 Marking.- Unit packages shall be marked in accordance with Standard MIL-STD-129 with the following information:

Stock number
Nomenclature
Quantity
Contract or order No.
Manufacturers' name and address

In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use.- Switchboard wattmeters are intended for use in general application of electrical measurement including shipboard and land installations. High-impact-shock-resistant wattmeters are preferred for shipboard use.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Complete type designation including voltage and current ratings (see 1.2.1).
(c) Dimensions required (see 3.5.1).
(d) External accessories, if required (see 3.6).
(e) Loop resistance of leads for external shunt of wattmeter (see 3.2.5.1).
(f) Whether for domestic shipment - immediate use: domestic shipment and storage, or overseas shipment (see 5.1 and 5.2).

6.2.1 Indirect shipments.- The packaging, packing, and marking requirements specified in section 5 apply only to direct purchases by or direct shipment to the Government and are not intended to apply to contracts or orders between the manufacturer and prime contractor.

6.3 Qualification.- In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on the Qualified Products List. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief of the Bureau of Ships, Department of the Navy, Washington 25, D. C.

6.4 Precautions.- Attention of equipment designers and manufacturers is invited to the fact that sufficient space should be provided in equipment for mounting standard meters of the maximum depth specified herein even though a shallower meter may be in the initial design.

6.4.1 Red mark on dial.- When specified in the contract or order, the manufacturer may place red marks on wattmeter dials at designated positions. There will be no change in the type designation to distinguish such wattmeters from those having normal markings; consequently, a wattmeter supplied for a certain application may have a red mark at an undesired position. If this occurs, it is recommended that the user remove or cover the red mark by any convenient method, and make other red marks if desired.

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6.5 This specification replaces that part of Specification 17M25 covering wattmeters.

Patent notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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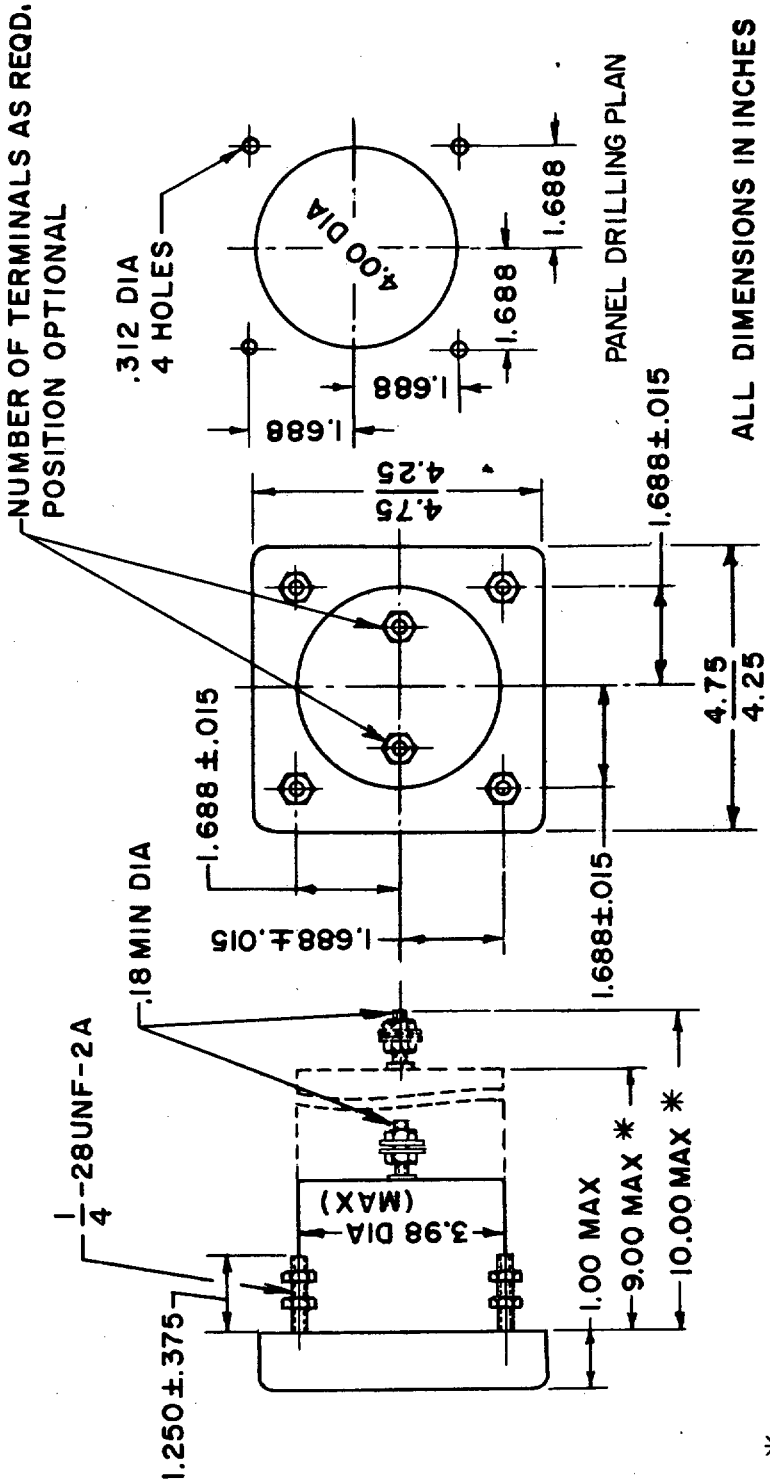


Figure 1.–Wattmeters, switchboard $4\frac{1}{2}$ inch, 250 degree scale.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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